Hans Yang

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Technical Skills

Languages: C, C++, HTML/CSS, Go, Java, JavaScript, Luau, Python, R, SQL, TypeScript, VBA

Software & Tech: AWS Lambda, AWS Sam, Azure, Docker, Excel/Google Sheets, Git, GitHub, Matplotlib, MongoDB, MySQL, Node.js, NumPy, Pandas, Redis, Vercel, Vite

Work Experience

Government of Ontario — 2026

Application Analyst/Developer @ MPBSDP, DevOps — *Hybrid*

- Debugged React components and enhanced TypeScript-based UI functionality using HTML, CSS, and SCSS.
- Used AI to document legacy code and workflows, streamlining maintenance and future debugging.
- Reproduced, diagnosed, and resolved software defects through targeted code fixes and application debugging.
- Collaborated with developers and QA through Microsoft Azure, using pull requests, bug reports, and issue tracking to maintain application quality.

Me & You Studios — 2026

Fullstack Gameplay Programmer, LiveOps — *Virtual*

- Engineered backend systems for player data persistence and implemented behavior trees for NPC decision-making, delivering new gameplay features coinciding with a **peak of 10,000+ concurrent users**.
- Developed frontend UI and VFX, contributing to the player-facing experience for a title that **scaled to 10K+ CCU.**
- Sourced, vetted, and onboarded external contractors, negotiating pricing and scope to align project needs and budget.
- Shipped content under tight deadlines with unit and integration test coverage, **boosting D1 retention by over 10%**
- Leveraged AI to improve documentation and navigate a complex legacy codebase, creating a more productive and maintainable development environment.

Projects

Visulie | *Luau, Roblox Studio, State/Data-Driven Architecture* — **2021 – Current**

- Designed and shipped a full-stack authoring tool with a custom editor, versioned data format, and a schema covering 900+ property/instance types, replacing thousands of lines of manual scripting with a declarative, data-driven workflow, sustained across 20+ releases as a commercial product with **150+ sales and $1.5K+ CAD revenue.**
- **Actively ranked #1 on Google Search**, increasing plugin visibility and outreach through marketplace optimization and community engagement.
- Profiled and optimized animation runtime performance through strategic caching, cache-hit optimization, and state-dependent algorithms to reduce redundant computation and improve execution efficiency.

AI Flashcard Study Tool | *AWS Lambda, AWS SAM, Docker, Gemini API, Go, Node.js, Typescript* — **2026 – Current**

- Designed a full-stack serverless AI document-processing pipeline using **AWS Lambda**, **AWS SAM**, and **Docker** to transform uploaded files into structured flashcard data through automated text extraction and generation workflows.
- Developed backend services using **Go** and client integrations with **TypeScript/Node.js**, establishing an end-to-end architecture for document ingestion, AI processing, and user-facing content delivery.

Fruit Picking Simulator (FPS) | *Luau, Roblox Studio* — **2023**

- Designed and shipped a full-stack interactive application with state-driven progression systems and a custom in-app economy, scaling to 100K+ users and 100+ CCU through iterative performance tuning and UX-driven front-end design.
- **Featured in the official industry year-in-review** for applying parallel/concurrent execution techniques Luau and Canvas Groups with awareness of parallel execution constraints to improve performance and rendering throughput.

Pocket Rift | *Luau, Roblox Studio, State Machine/Event/Authoritative Server Arcitecture* — **2019 – Current**

- Implemented a product analytics pipeline to track user funnels and onboarding behavior, enabling data-driven iteration that improved conversion rates by over 185%.
- Designed a distributed client-server networking architecture with client prediction, state synchronization, and delta compression, **reducing latency and network bandwidth usage by 95% over baseline implementations.**
- Built reusable internal frameworks, including an in-memory caching system for session data and a reactive state management system, reducing system coupling and enabling scalable client-server architecture.

Education

Wilfrid Laurier University — Sep. 2024 – Present

Bachelor of Science (Computer Science) — *Major GPA: 11.46/12 (88%)*

Relevant Courses: Algorithm Design & Analysis, Calculus I & II, Computer Networks, Databases I, Software Engineering, Data Structures I & II, Introduction to Microprocessors, Discrete Mathematics, Object Oriented Programming, Probability I